F - Strongly Connected 2

分值：525 分

Problem Statement

There is a directed graph with N vertices and $N-1+M$ edges, with edges and vertices numbered.
For $1 \leq i \leq M$, edge i is a directed edge from vertex X_i to vertex Y_i, and for $1 \leq i \leq N-1$, edge $M+i$ is a directed edge from vertex $i+1$ to vertex i.

有一个包含 N 个顶点和 $N-1+M$ 条边的有向图，边和顶点均已编号。 对于 $1 \leq i \leq M$，边 i 是从顶点 X_i 到顶点 Y_i 的有向边；对于 $1 \leq i \leq N-1$，边 $M+i$ 是从顶点 $i+1$ 到顶点 i 的有向边。

There are 2^M ways to choose some (possibly zero) edges from among edges $1, 2, \ldots, M$. Among these ways, how many result in the graph being strongly connected after deleting the chosen edges? Find the count modulo 998244353.

从边 $1, 2, \ldots, M$ 中选择若干条（可以是 0 条），共有 2^M 种选择方式。在这些方式中，有多少种方式使得删除选中的边后，图仍然是强连通的？求答案对 998244353 取模后的结果。

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 2 \times 10^5$
- $1 \leq X_i < Y_i \leq N$
- All input values are integers.
- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 2 \times 10^5$
- $1 \leq X_i < Y_i \leq N$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入以如下格式从标准输入给出：

$$
\begin{array}{l}
N\ M \\
X_1\ Y_1 \\
X_2\ Y_2 \\
\vdots \\
X_M\ Y_M
\end{array}
$$

Output

Output the answer.

输出答案。

Sample Input 1

```
4 3
1 4
1 3
2 4
```

Sample Output 1

```
5
```

The graph is strongly connected when the set of chosen edge indices is one of $\{\}, \{1\}, \{2\}, \{3\}, \{2,3\}$, giving five ways.

当选中的边的下标集合为 $\{\}, \{1\}, \{2\}, \{3\}, \{2,3\}$ 中的任意一个时，图都是强连通的，因此共有 5 种方式。

Sample Input 2

```
10 11
1 4
1 4
3 9
```

```
2 5
3 4
9 10
6 9
4 10
1 3
8 10
4 7
```

Sample Output 2

```
1297
```

The given graph may have multi-edges.

给定的图可能存在重边。